

Class 11: Protein Prediction With AlphaFold

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Overview

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence.

Querying the AlphaFold database

My top hit was Gag-Pol polyprotein from HIV-1 with a high confidence score of pLDDT 96.78)

Custom Analysis of resulting models

```
results_dir <- "HIV_Pr_Dimer_23119"

pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)

basename(pdb_files)
```

```
[1] "HIV_Pr_Dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "HIV_Pr_Dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "HIV_Pr_Dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "HIV_Pr_Dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "HIV_Pr_Dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)
```

```
# Read all data from Models
# and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
HIV_Pr_Dimer_23119/HIV_Pr_Dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
HIV_Pr_Dimer_23119/HIV_Pr_Dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
HIV_Pr_Dimer_23119/HIV_Pr_Dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
HIV_Pr_Dimer_23119/HIV_Pr_Dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
HIV_Pr_Dimer_23119/HIV_Pr_Dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....
```

Extracting sequences

```
pdb/seq: 1    name: HIV_Pr_Dimer_23119/HIV_Pr_Dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
pdb/seq: 2    name: HIV_Pr_Dimer_23119/HIV_Pr_Dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
pdb/seq: 3    name: HIV_Pr_Dimer_23119/HIV_Pr_Dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
pdb/seq: 4    name: HIV_Pr_Dimer_23119/HIV_Pr_Dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
pdb/seq: 5    name: HIV_Pr_Dimer_23119/HIV_Pr_Dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
```

```
pdbs
```

```

1          .          .          .          .          50
[Truncated_Name:1]HIV_Pr_Dim PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]HIV_Pr_Dim PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]HIV_Pr_Dim PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]HIV_Pr_Dim PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]HIV_Pr_Dim PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
*****
1          .          .          .          .          50

51          .          .          .          .          100
[Truncated_Name:1]HIV_Pr_Dim GGFIVRQYDQILIEICGHKAIGTVLVGPTPVNIIGNLLTQIGCTLNFP
```

```

[Truncated_Name:2]HIV_Pr_Dim  GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]HIV_Pr_Dim  GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]HIV_Pr_Dim  GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]HIV_Pr_Dim  GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
*****
51      .      .      .      .      100

101      .      .      .      .      150
[Truncated_Name:1]HIV_Pr_Dim  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:2]HIV_Pr_Dim  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:3]HIV_Pr_Dim  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:4]HIV_Pr_Dim  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:5]HIV_Pr_Dim  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
*****
101      .      .      .      .      150

151      .      .      .      .      198
[Truncated_Name:1]HIV_Pr_Dim  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]HIV_Pr_Dim  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]HIV_Pr_Dim  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]HIV_Pr_Dim  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]HIV_Pr_Dim  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
*****
151      .      .      .      .      198

```

Call:

```
pdbaln(files = pdb_files, fit = TRUE, exe_file = "msa")
```

Class:

```
pdbs, fasta
```

Alignment dimensions:

```
5 sequence rows; 198 position columns (198 non-gap, 0 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

```
rd <- rmsd(pdb, fit=T)
```

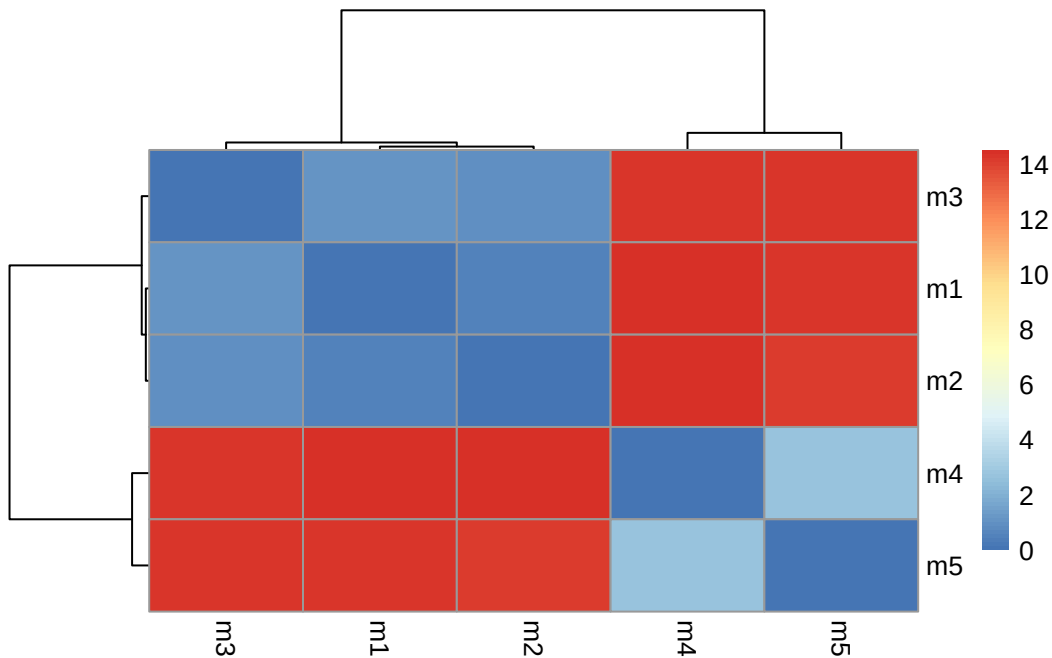
Warning in rmsd(pdb, fit = T): No indices provided, using the 198 non NA positions

```
range(rd)
```

```
[1] 0.000 14.539
```

```
library(pheatmap)
```

```
colnames(rd) <- paste0("m",1:5)  
rownames(rd) <- paste0("m",1:5)  
pheatmap(rd)
```



My RMSD heatmap shows that some predicted models are more similar to each other than others. The blue squares represent lower RMSD values, meaning those model pairs overlap more closely, while the red squares represent higher RMSD values, meaning those models are more structurally different. In my heatmap, models 1 and 2 appear more similar to each other, while models 3, 4, and 5 are more similar as a separate group. This suggests that the five AlphaFold predictions are mostly consistent but still have some variation between model groups.

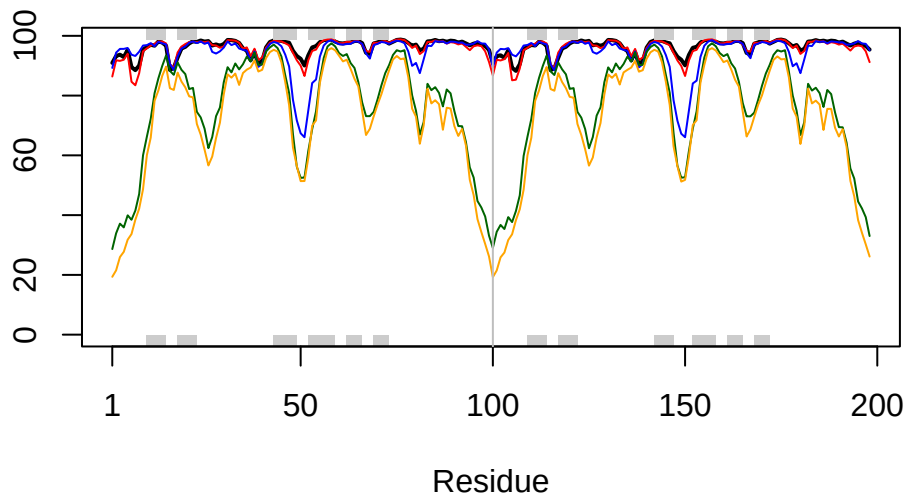
```
# Read a reference PDB structure  
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```

plotb3(pdbb$b[1,], typ="l", lwd=2, sse=pdbb)
points(pdbb$b[2,], typ="l", col="red")
points(pdbb$b[3,], typ="l", col="blue")
points(pdbb$b[4,], typ="l", col="darkgreen")
points(pdbb$b[5,], typ="l", col="orange")
abline(v=100, col="gray")

```



```

core <- core.find(pdbb)

```

```

core size 197 of 198  vol = 9757.214
core size 196 of 198  vol = 8985.811
core size 195 of 198  vol = 3169.051
core size 194 of 198  vol = 1572.467
core size 193 of 198  vol = 1415.614
core size 192 of 198  vol = 1352.407
core size 191 of 198  vol = 1291.262
core size 190 of 198  vol = 1236.417
core size 189 of 198  vol = 1145.775
core size 188 of 198  vol = 1091.912
core size 187 of 198  vol = 1034.344
core size 186 of 198  vol = 982.07
core size 185 of 198  vol = 929.556

```

core size 184 of 198	vol = 870.277
core size 183 of 198	vol = 832.754
core size 182 of 198	vol = 793.235
core size 181 of 198	vol = 759.982
core size 180 of 198	vol = 725.279
core size 179 of 198	vol = 684.372
core size 178 of 198	vol = 657.54
core size 177 of 198	vol = 635.321
core size 176 of 198	vol = 615.026
core size 175 of 198	vol = 595.048
core size 174 of 198	vol = 574.548
core size 173 of 198	vol = 558.08
core size 172 of 198	vol = 530.216
core size 171 of 198	vol = 515.328
core size 170 of 198	vol = 500.325
core size 169 of 198	vol = 481.989
core size 168 of 198	vol = 463.138
core size 167 of 198	vol = 447.958
core size 166 of 198	vol = 433.407
core size 165 of 198	vol = 418.55
core size 164 of 198	vol = 407.611
core size 163 of 198	vol = 388.191
core size 162 of 198	vol = 375.704
core size 161 of 198	vol = 364.118
core size 160 of 198	vol = 355.318
core size 159 of 198	vol = 342.769
core size 158 of 198	vol = 330.932
core size 157 of 198	vol = 319.902
core size 156 of 198	vol = 312.329
core size 155 of 198	vol = 301.544
core size 154 of 198	vol = 292.043
core size 153 of 198	vol = 277.975
core size 152 of 198	vol = 264.567
core size 151 of 198	vol = 259.108
core size 150 of 198	vol = 249.631
core size 149 of 198	vol = 238.54
core size 148 of 198	vol = 228.057
core size 147 of 198	vol = 219.476
core size 146 of 198	vol = 209.149
core size 145 of 198	vol = 200.309
core size 144 of 198	vol = 191.697
core size 143 of 198	vol = 181.697
core size 142 of 198	vol = 172.238

core size 141 of 198 vol = 162.501
core size 140 of 198 vol = 152.814
core size 139 of 198 vol = 144.214
core size 138 of 198 vol = 134.532
core size 137 of 198 vol = 129.459
core size 136 of 198 vol = 126.555
core size 135 of 198 vol = 120.409
core size 134 of 198 vol = 114.556
core size 133 of 198 vol = 108.739
core size 132 of 198 vol = 105.076
core size 131 of 198 vol = 99.07
core size 130 of 198 vol = 94.654
core size 129 of 198 vol = 90.868
core size 128 of 198 vol = 87.652
core size 127 of 198 vol = 83.72
core size 126 of 198 vol = 80.157
core size 125 of 198 vol = 75.296
core size 124 of 198 vol = 69.567
core size 123 of 198 vol = 66.175
core size 122 of 198 vol = 63.445
core size 121 of 198 vol = 60.166
core size 120 of 198 vol = 56.825
core size 119 of 198 vol = 53.932
core size 118 of 198 vol = 51.094
core size 117 of 198 vol = 49.028
core size 116 of 198 vol = 45.741
core size 115 of 198 vol = 41.941
core size 114 of 198 vol = 38.856
core size 113 of 198 vol = 36.504
core size 112 of 198 vol = 33.827
core size 111 of 198 vol = 32.108
core size 110 of 198 vol = 29.123
core size 109 of 198 vol = 27.622
core size 108 of 198 vol = 26.85
core size 107 of 198 vol = 25.83
core size 106 of 198 vol = 24.165
core size 105 of 198 vol = 22.397
core size 104 of 198 vol = 21.963
core size 103 of 198 vol = 22.078
core size 102 of 198 vol = 19.773
core size 101 of 198 vol = 17.479
core size 100 of 198 vol = 15.947
core size 99 of 198 vol = 13.538

```

core size 98 of 198  vol = 11.818
core size 97 of 198  vol = 10.202
core size 96 of 198  vol = 8.599
core size 95 of 198  vol = 7.179
core size 94 of 198  vol = 5.863
core size 93 of 198  vol = 4.292
core size 92 of 198  vol = 3.546
core size 91 of 198  vol = 3.013
core size 90 of 198  vol = 2.427
core size 89 of 198  vol = 1.635
core size 88 of 198  vol = 1.063
core size 87 of 198  vol = 0.755
core size 86 of 198  vol = 0.641
core size 85 of 198  vol = 0.552
core size 84 of 198  vol = 0.483
FINISHED: Min vol ( 0.5 ) reached

```

```
core.inds <- print(core, vol=0.5)
```

```

# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1      8  48     41
2     52  95     44

```

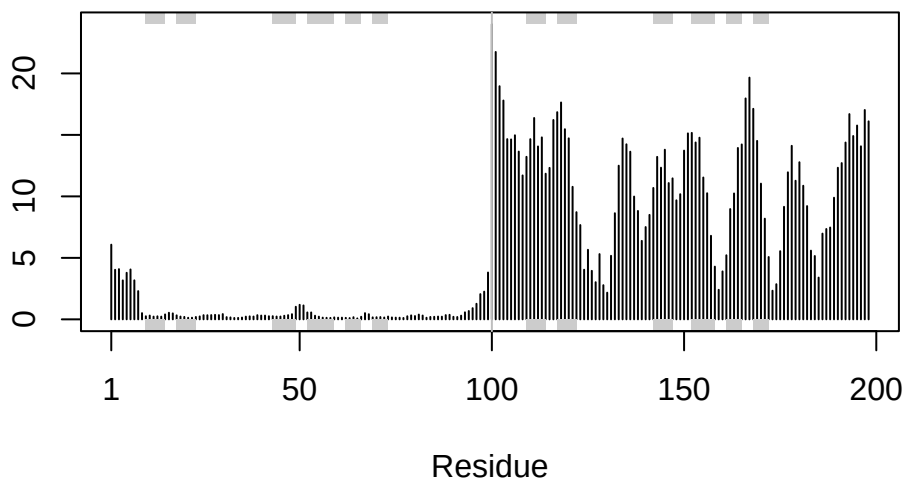
```
xyz <- pdbfit(pdb, core.inds, outpath="corefit_structures")
```

```
rf <- rmsf(xyz)
```

```

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")

```

Predicted Alignment Error for Domains

```
library(jsonlite)
# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"  "max_pae" "pae"      "ptm"      "iptm"
```

```
# Per-residue pLDDT scores
# same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 90.94 93.25 93.69 92.88 95.31 89.50
```

Maximum Pae values

```
pae1$max_pae
```

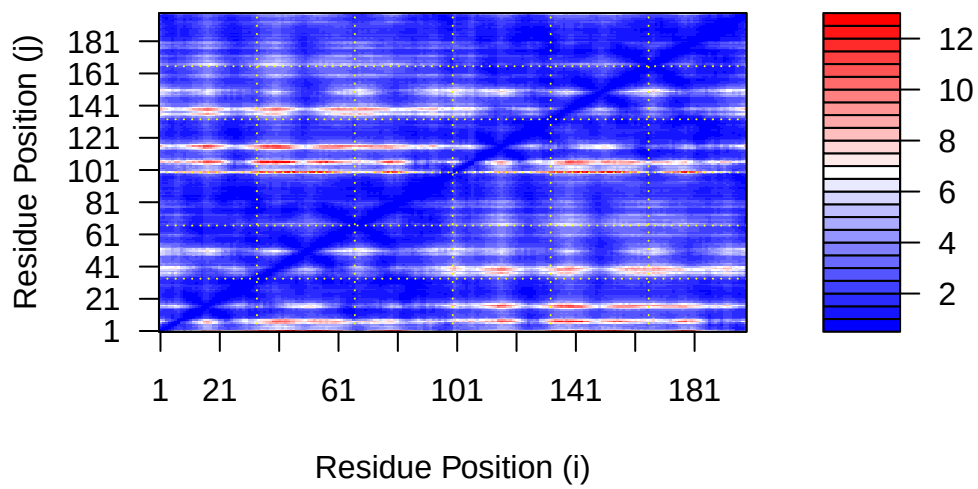
```
[1] 12.73438
```

```
pae5$max_pae
```

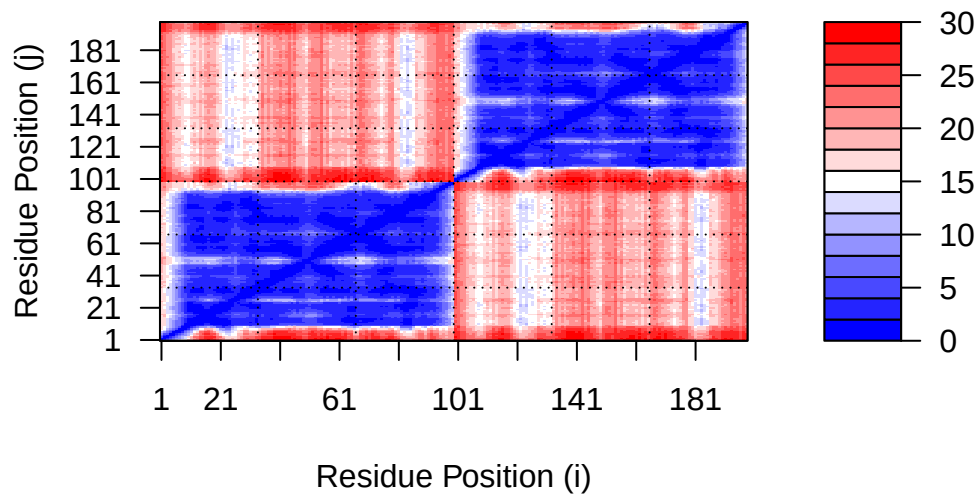
```
[1] 29.92188
```

Plot N by N

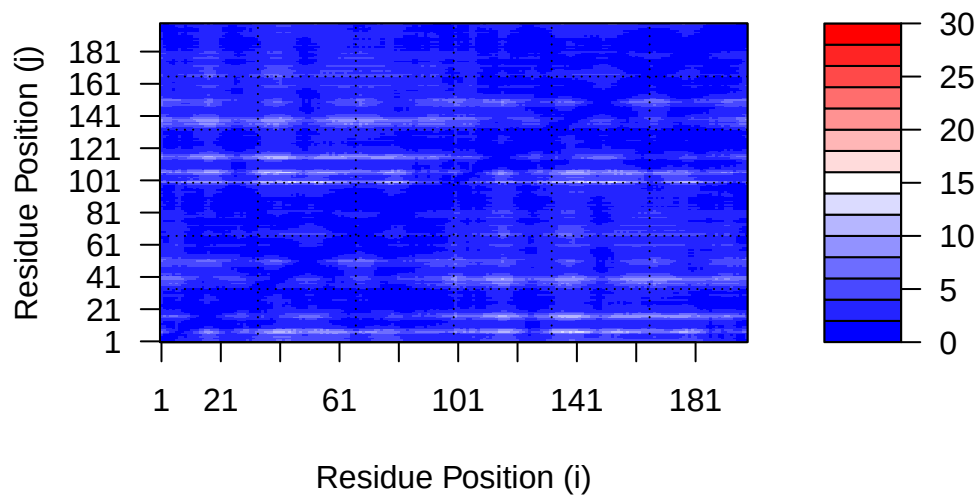
```
plot.dmat(pae1$pae,  
          xlab="Residue Position (i)",  
          ylab="Residue Position (j)")
```



```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue Conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                      pattern=".a3m$",
                      full.names = TRUE)
aln_file
```

```
[1] "HIV_Pr_Dimer_23119/HIV_Pr_Dimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
```

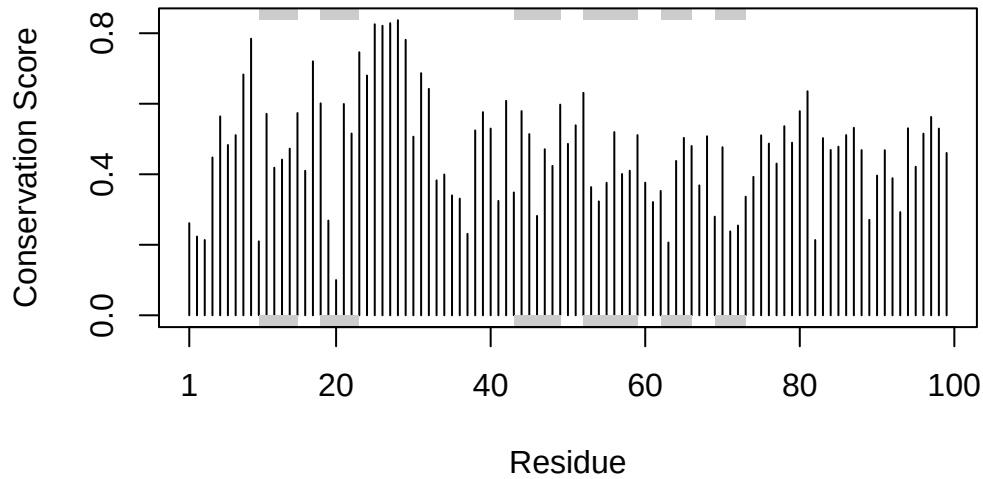
```
[2] " ** Duplicated sequence id's: 101 **"
```

How many sequences are in this alignment

```
dim(aln$ali)
```

```
[1] 5397 132
```

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```